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Question:	1	2	3	4	Total
Points:	2½	2½	2½	2½	10
Score:	2.25	2	2.5	1	7.75

Everything must be justified.

1. Consider the following ordinary differential equation

$$\frac{dx}{dt} = 3x(t)\cos(t) \qquad \textbf{para } t > 0, \tag{1}$$

with $x(0) = 1$.

B (a) (1 point) Use Taylor method of order 2 to solve numerically the differential equation for $t \in [0, 0.2]$. Use $\Delta t = 0.10$. Fill in the table with five digits after the decimal point.

B (b) (½ point) Repeat the previous exercise using $\Delta t = 0.05$.

B (c) (1 point) Estimate the step Δt necessary to get a final global error smaller than 10^{-6}.

t_k	$x_k^{(a)}$ $\Delta t = 0.10$	$x_k^{(b)}$ $\Delta t = 0.05$
0.00	1.00000	1.00000
0.05	XXXXXXXXXXXXXX	1.16125
0.10	1.34500	1.34803
0.15	XXXXXXXXXXXXXX	1.56374
0.20	1.80439	1.81199

2. Consider the least squares problem of adjusting data in the following table to the function $y = \alpha x^2 + \beta$.

x_k	y_k
-1	+1
+0	+2
+1	+3

M (a) (½ point) Explain clearly why the problem can be solved using QR decomposition.

B (b) (1½ points) Solve using QR decomposition.

B (c) (½ point) Find the quadratic error without computing $\|\mathbf{A}\vec{x} - \vec{b}\|_2^2$.

3. (2½ points) Consider the problem of finding the minimum of the function B

$$f(x, y) = (x - y)^2 + (y - 2)^2. \tag{2}$$

Starting with $(x_0, y_0) = (0, 0)$, do two iterations of the Newton algorithm. Fill in the following table with your results. Use five digits after the decimal point.

k	x_k	y_k
0	0.00000	0.00000
1	2.00000	2.00000
2	2.00000	2.00000

4. (a) (½ point) Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & +10 \\ +10 & -2 \\ +1 & +2 \end{pmatrix}. \tag{3}$$

How many positive singular values does $\mathbf{A}$ have? Justify your answer.

(b) (1 point) Calculate $\|(\mathbf{A})\|_2$.

(c) (1 point) Find the Lagrange-Chebychev interpolating polynomial $p_2(x) = a_2x^2 + a_1x + a_0$ for the function $g(x) = \ln(1+x)$ in $[-0.1, +0.1]$. Fill in the following table with five digits after the decimal point.

a_2	a_1	a_0
-0.92420	1.52069	0.00000

Wrong